

LAB PROGRESS REPORT · 2 OCTOBER 2026 · VERSION 4

# Evaluating in-silico perturbation with Geneformer

Tumour-infiltrating T cells, the criteria a result must meet,  
and a first look at transposable-element regulation

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# Question: which genes, when perturbed in silico, shift tumour-infiltrating T cells toward the normal-tissue state?

T cell in a  
lung tumour

perturb one gene



in the model

T cell in the same  
donor's normal lung

T cells infiltrating a tumour often differ in state from T cells in the same donor's normal tissue. Genes that maintain the tumour-associated state are candidate targets. Testing each gene experimentally is slow, so we use a pretrained model to prioritise genes, and we ask how far its predictions can be trusted.

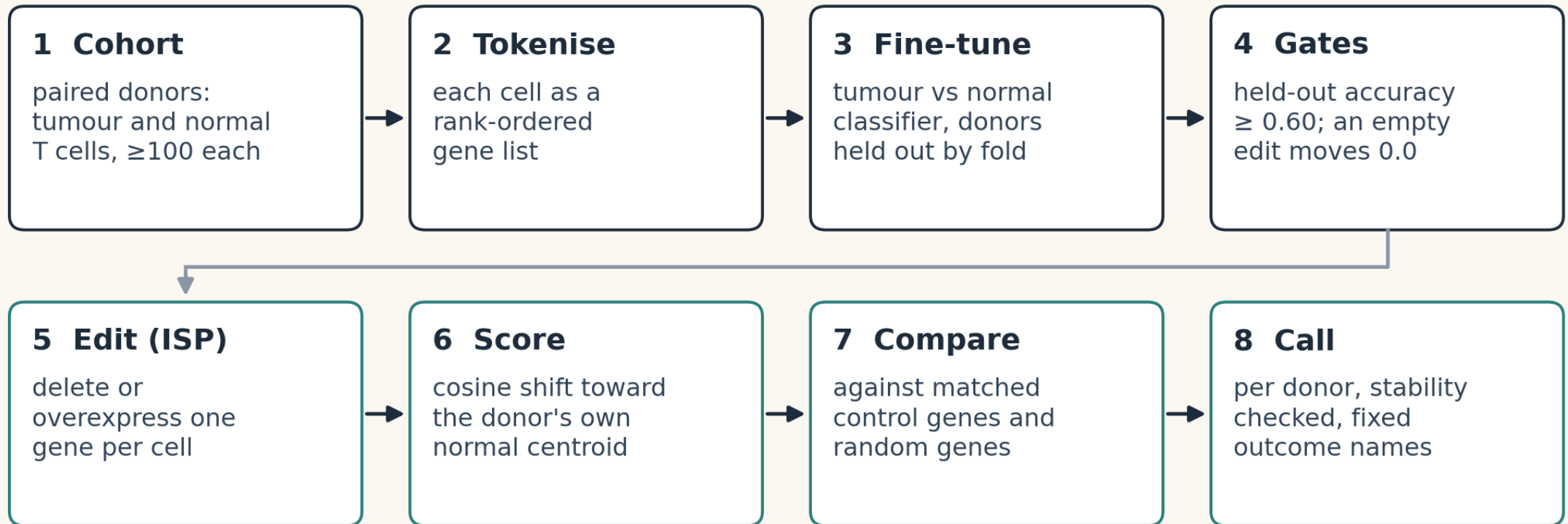
# Outline

1. **Method.** The analysis pipeline, rank-value tokenisation, fine-tuning and in-silico perturbation.
2. **Lessons.** Four results from July to September that did not hold, or held only in part.
3. **Criteria.** Six checks we now apply, each traced to one of those lessons.
4. **Status.** External cohorts, the colon study and what the evidence supports.
5. **Bulk RNA-seq.** A first test of perturbation prediction on bulk profiles, against measured knockouts.
6. **Transposable elements.** The same question for TE regulators: design, two fish analyses, and two ground-truth tests (zebrafish, T-cell knockouts).
7. **Next.** Criteria, open questions and the outline of a methods paper.

Colour code: ✓ passed ✗ failed ? not yet known

# The analysis runs in eight steps, from a paired cohort to a donor-level call

Diagram



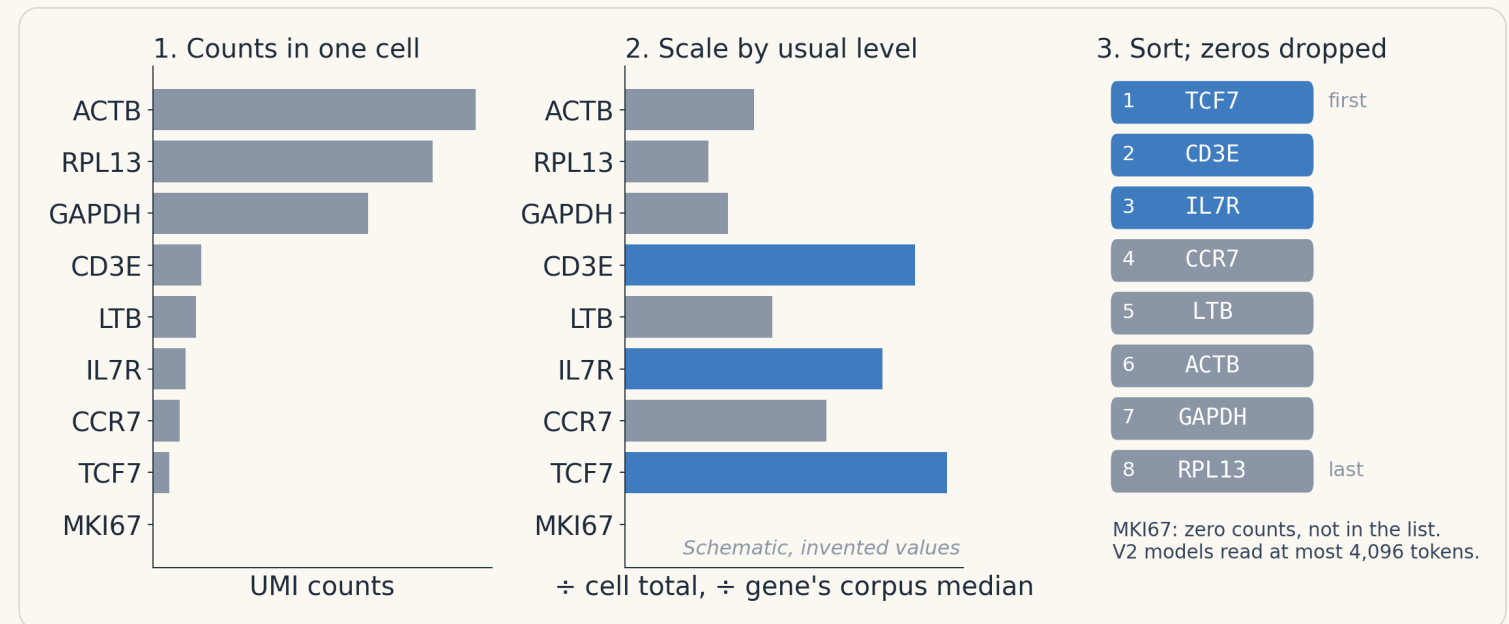
Steps 1 to 4 build and test the instrument; no perturbation is run unless both gates pass. Steps 5 to 8 perturb, score and judge each gene.

# Geneformer encodes each cell as a list of genes ranked by relative expression

Counts are divided by the cell's total, then by each gene's median level across the pretraining corpus, and sorted. Housekeeping genes fall; genes unusually high in this cell rise.

Genes with zero counts are left out. The ranked list is the model's input; each gene is a **token**.

*V2 models accept at most 4,096 tokens per cell.*



# The model was pretrained on about 104 million cells by predicting masked genes

CD3E

IL7R

?

LTB

TCF7

...

Pretraining task: mask a gene and predict it from the rest of the list. Illustration, not data.

After pretraining, the model maps each cell to a vector, its **embedding**. Cells with similar ranked lists have similar embeddings.

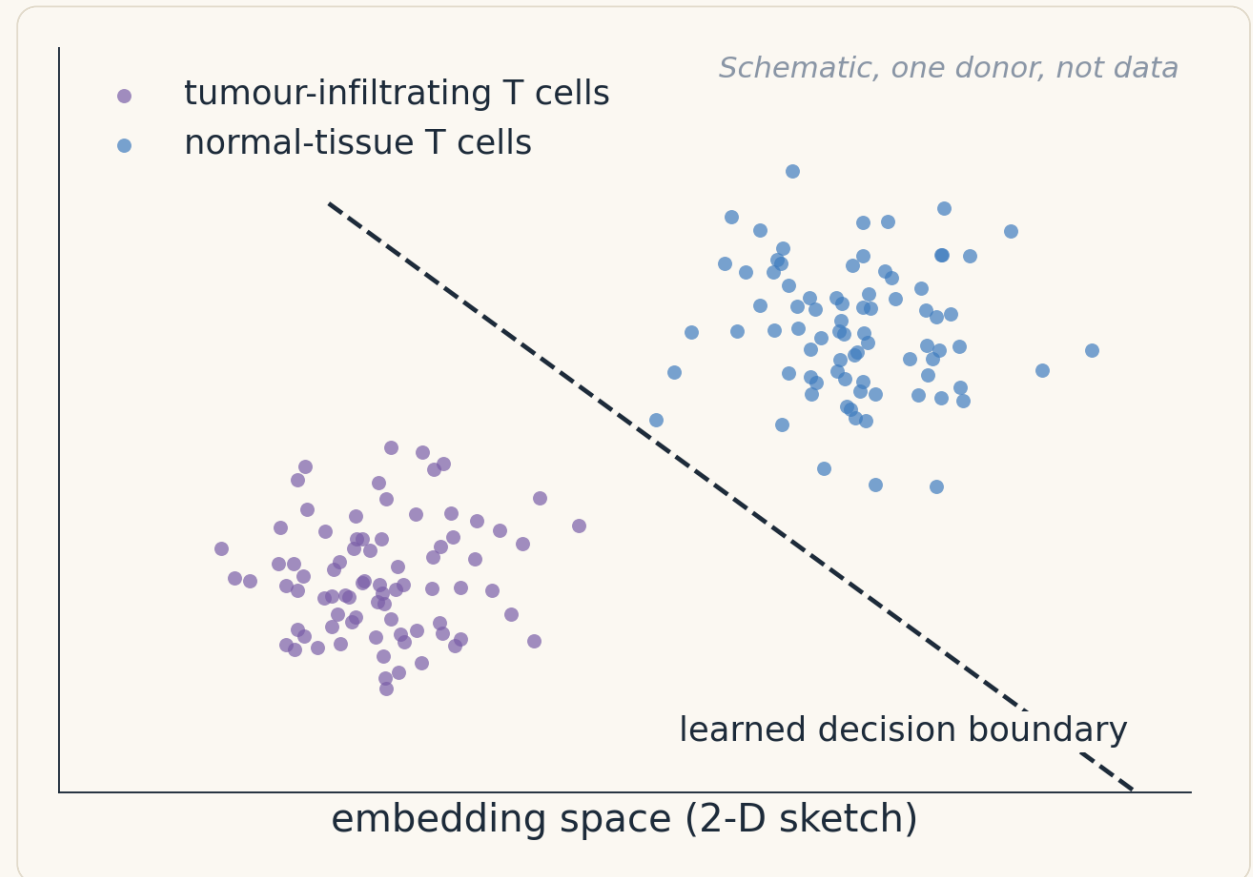
*We have used two model sizes; the larger is used for all results shown here unless stated.*

# Fine-tuning adds one task: separating a donor's tumour T cells from normal T cells

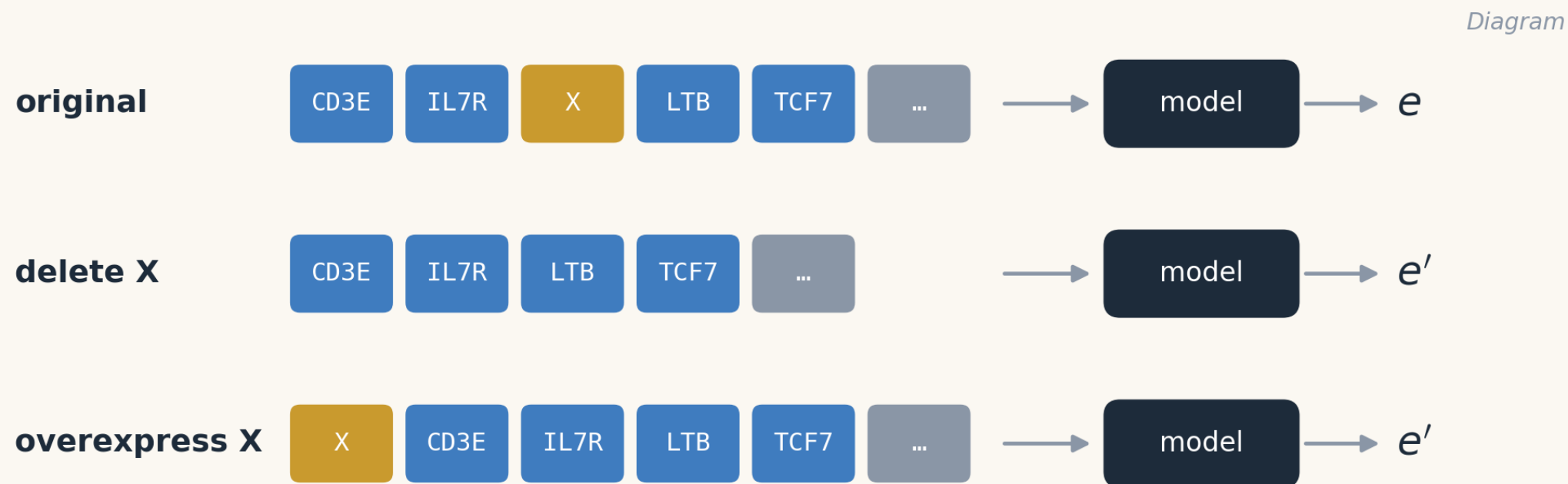
The classifier is trained on some donors and scored only on donors it never saw (donor-level cross-fitting).

Lung (LUAD, 43 donors): pooled held-out balanced accuracy **0.825**; all **43 of 43** donors above chance (0.5).

*Balanced accuracy weights tumour and normal cells equally.*



# An in-silico perturbation edits the token list and measures how the embedding moves



Score per cell:  $\cos(e', g) - \cos(e, g)$ ,  $g$  = centroid of the same donor's normal T cells.

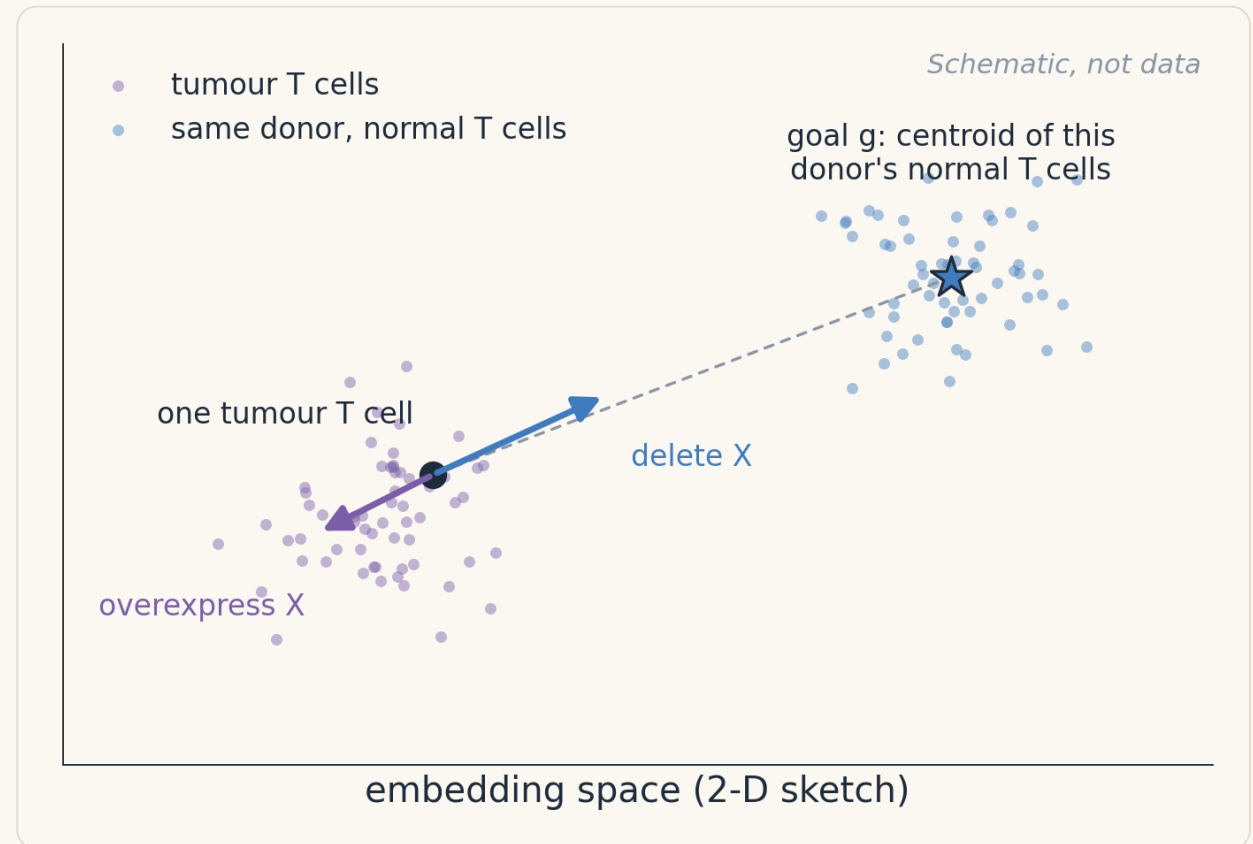
Mean over the donor's cells that carry X, then median over donors ( $\geq 10$  donors with  $\geq 10$  such cells).

# The score is a movement in the model's embedding space, not a measured biological effect

A positive score means the edited cell moved toward the donor's normal centroid in the model's representation.

Whether a real knockout or overexpression would do the same is a separate question. Answering it needs experimental data.

*No cell is modified. The model always returns a score, whether or not the gene is relevant.*



# The first results, in July and August, appeared encouraging

## July: genome-wide deletion screen

21,000 lung T cells, the smaller model,  
2,937,776 single-gene deletions on held-out  
cells.

*Internal audit, 28 July: the adenocarcinoma and  
squamous-cell analyses could not on their own support  
the conference abstract's claims about small-cell lung  
cancer.*

## August: conference talk

A 50-gene panel in small-cell lung cancer  
(SCLC). Four genes passed in both directions in  
all three test donors: deletion shifted cells  
toward normal, overexpression away.

TIM-3, TIGIT, CTLA-4 and IL7R, genes  
associated with T-cell exhaustion or persistence.

# Lesson 1: the strongest early hits were not T-cell signals

## Top of the screen

S100A8 and S100A9, among the most significant genes, shifted cells in the *same* direction under deletion and overexpression in 9 of 12 comparisons.

Of the top 120 results, 55 were flagged as likely **ambient RNA**, transcripts from other cells captured during sample preparation.

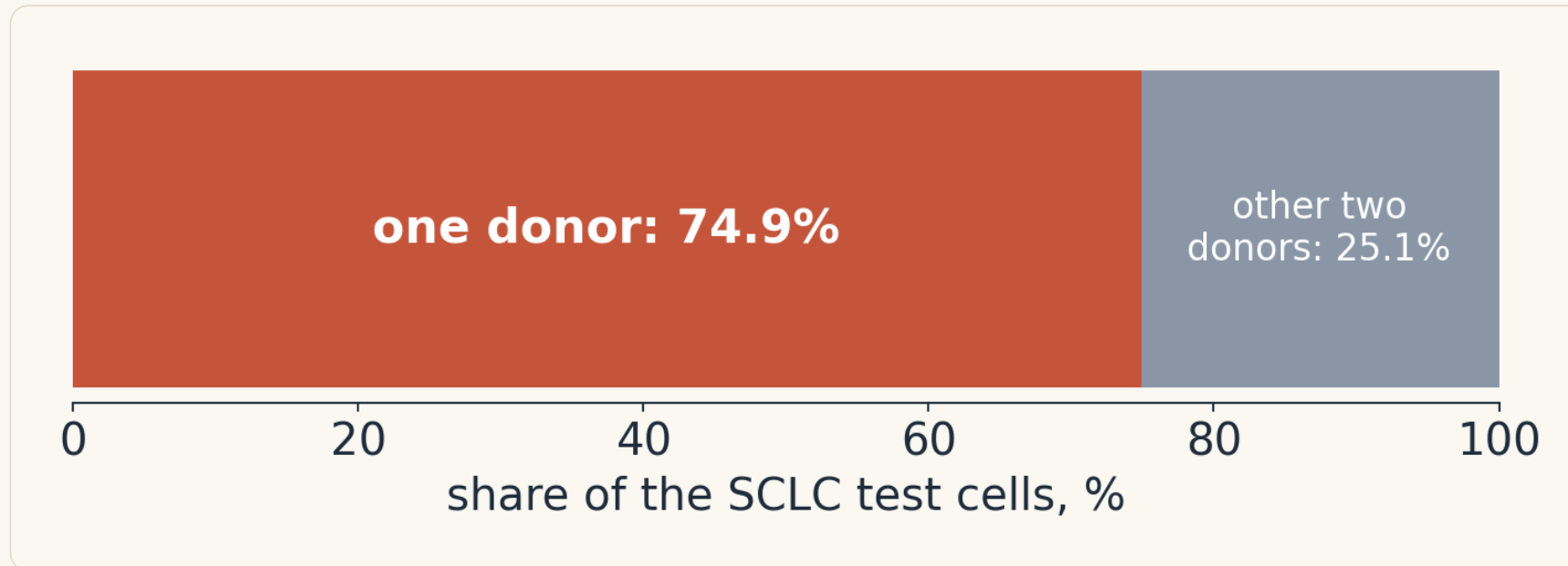
## The comparison groups

In July, tumour and normal T cells came from different studies; none of the 18 studies supplied both. The classifier could partly separate studies rather than tissues.

13 of the 15 July top genes could not be tested in T cells.

Addressed by  2  4

## Lesson 2: one donor can dominate a cell-level comparison



Two analyses appeared to disagree about small-cell lung cancer. One donor supplied 74.9% of the SCLC test cells. Counted by donor (19 against 22 donors), the difference was not significant:  $p = 0.216$ .

Addressed by  6

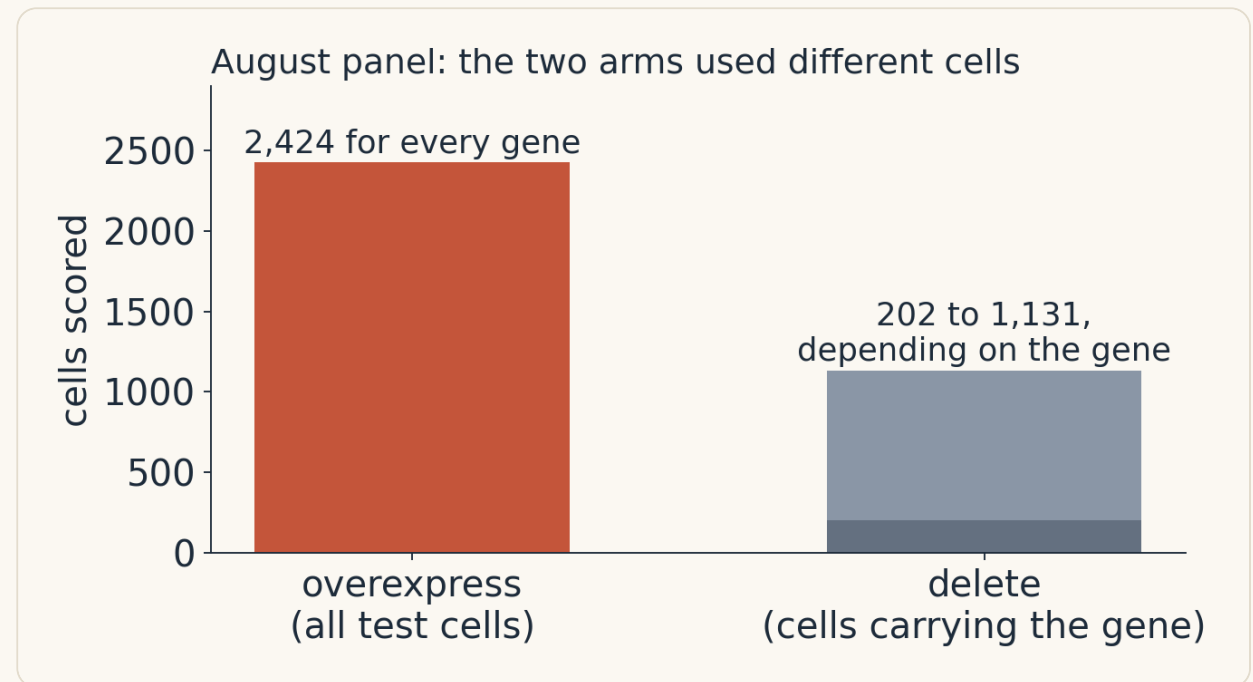
## Lesson 3: the two perturbation arms were scored on different cells

In the August panel, overexpression was scored on every test cell and deletion only on cells that expressed the gene.

The cause was a default in Geneformer's perturbation code: given a gene list, it inserts an overexpressed gene into every cell (14,738 of 15,179 calls in our rebuilt study).

*The August genes carry a qualifier; they were not retracted. The panel has not been rerun.*

Addressed by  3



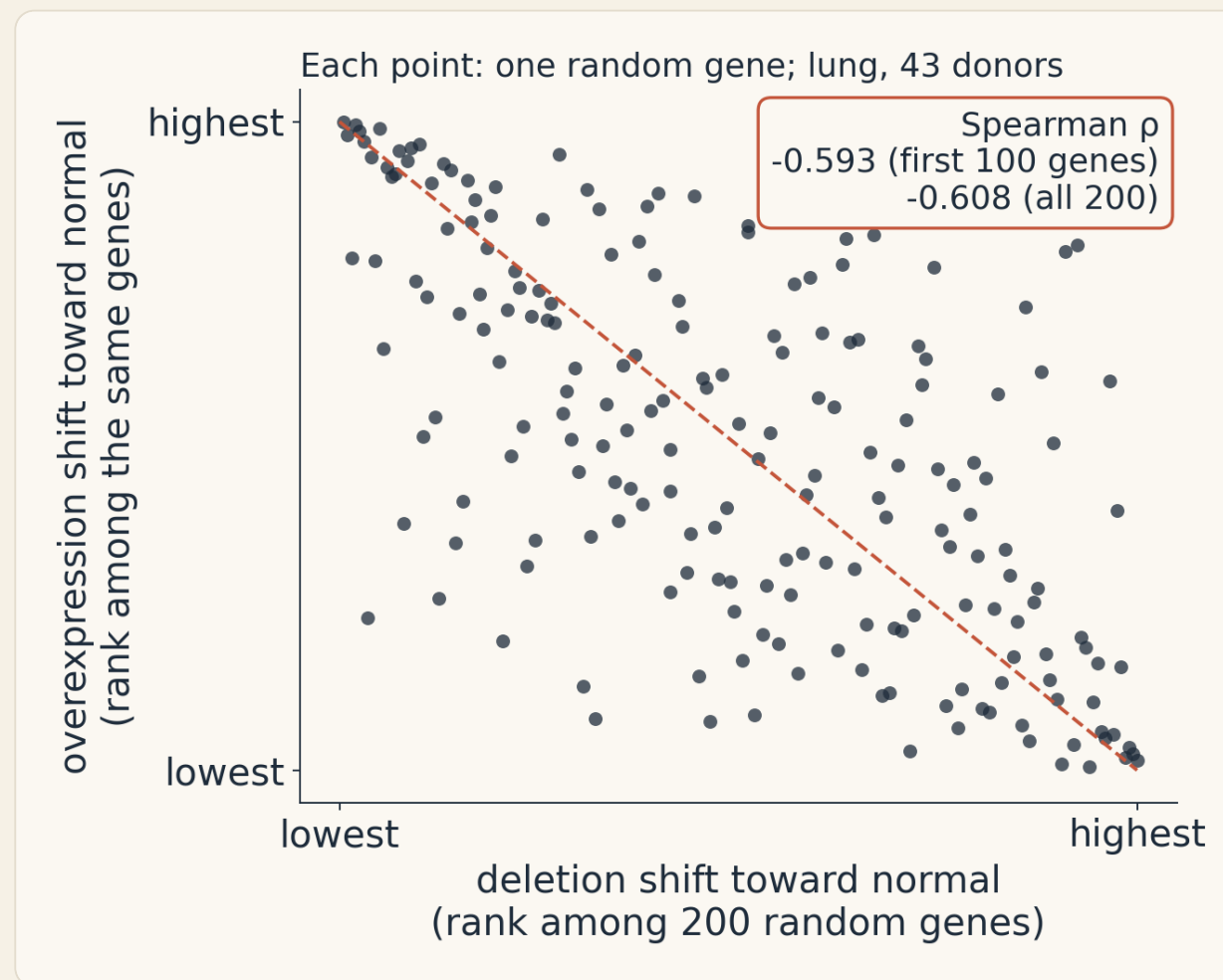
## Lesson 4: random genes show the opposed pattern we had treated as evidence

We ran the same pipeline on genes drawn at random from those detected in lung tumour T cells.

Deletion and overexpression shifts were anti-correlated: Spearman  $\rho = -0.593$  (100 genes) and  **$-0.608$**  (200 genes).

*One-sided permutation  $p$  at the test's floor, at most about 1 in 100,000.*

Addressed by  5



# Six criteria, applied in order, stand between a result and a claim

Diagram (ISP-STD-1)



Fixed outcome names (never "no effect"):



Adopted on 28 September 2026 as the lab's standard for judging in-silico perturbation (ISP-STD-1).

## 1 Register the test before any result exists

Before a run, we record the test, the predicted direction, the significance threshold and the wording for each outcome. An independent reviewer approves the exact files.

We also confirm that the test **can** reach significance with the expected sample size.

From: the August panel, where we interpreted an ordering we had not planned to test.

### EXAMPLE: THE COLON STUDY

10 lung reference genes are expected to be testable in colon. The lung pattern counts as repeated if at least **9 of 10** keep their lung direction (8 of 9 if only 9 are testable).

The first review of the plan did not pass: it described the software environment as unchanged, but one library had been downgraded. The plan was corrected before the run.

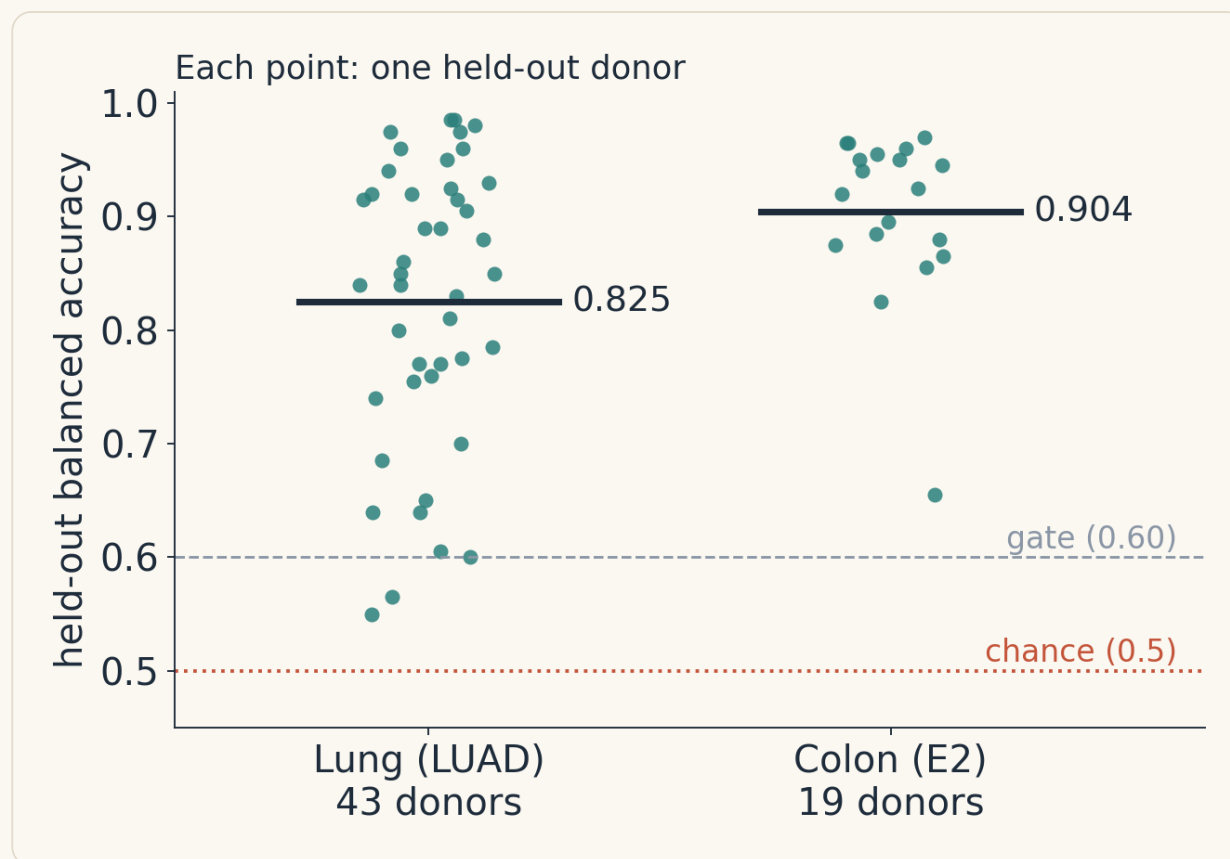
## 2 Confirm the instrument works before reading anything from it

**Classifier gate.** Held-out balanced accuracy  $\geq 0.60$  and more donors above chance than expected (sign test). Otherwise the study stops.

**No-op gate.** Two passes with no edit must give a shift of exactly zero.

Colon: ✓ 0.904; 19 of 19 donors ✓ no-op shift 0.0

From: the July classifier, which may have separated studies rather than tissues.



## ⊖ 3 Score both perturbations on exactly the same cells

### ✗ August panel

Overexpression: 2,424 cells for every gene.  
Deletion: 202 to 1,131 cells.

### ✓ Since 25 September

Both arms are analysed on the same cells, and every run checks the cell counts. (Panel code fixed 22 September; rebuilt study corrected 25 September.)

Diagnostic: if one arm's cell count is constant across genes while the other varies, the arms are not paired.

| From: lesson 3.

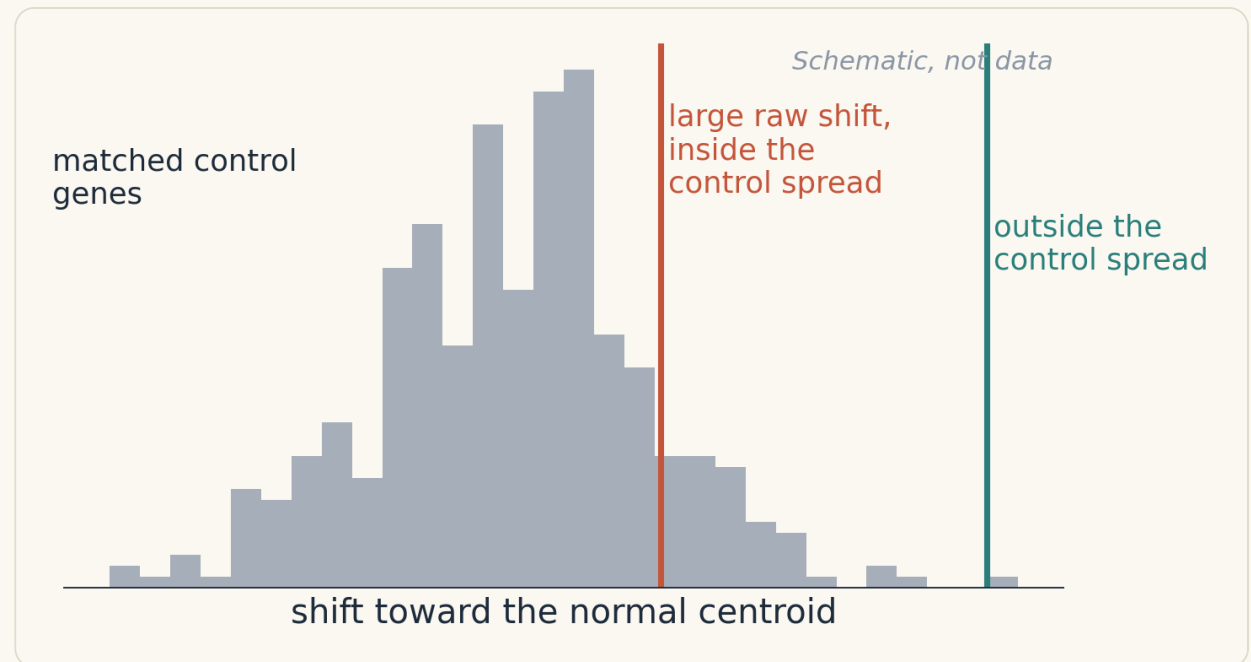
## ⚖️ 4 A gene must shift cells more than matched control genes

Widely detected genes shift cells more regardless of function. Each gene is compared with **at least 20 control genes** matched on detection rate and mean rank.

If fewer than 20 can be found, the gene is reported as *not estimable*; the matching tolerance is never widened.

Likely ambient-RNA genes and classifier-anchor genes are flagged.

| From: lesson 1 (S100A8, ambient RNA).



## 5 Opposed effects of the two perturbations are not, on their own, evidence

### What the random genes show

Opposed effects are what this model and design produce for genes in general.

The study's control genes agreed in sign:  $\rho = -0.658$  (308 of 318 genes with enough data).

### What they do not show

They do not remove the panel result: 13 of 34 testable T-cell genes still exceed their matched controls.

Evidence now means an unusually large effect relative to this baseline, not opposite signs.

From: lesson 4.

## 👤 6 Count donors, require stability, and name every outcome in advance

### Donors and stability

The donor, not the cell, is the unit of inference.

A result must hold when any single control gene is dropped, and in at least 95% of 10,000 bootstrap draws.

### Fixed outcome names

`positive` `negative` `opposite_direction`

`not_estimable` `..._open`

We do not write "no effect". A negative result means this test, at this sample size, did not meet its threshold.

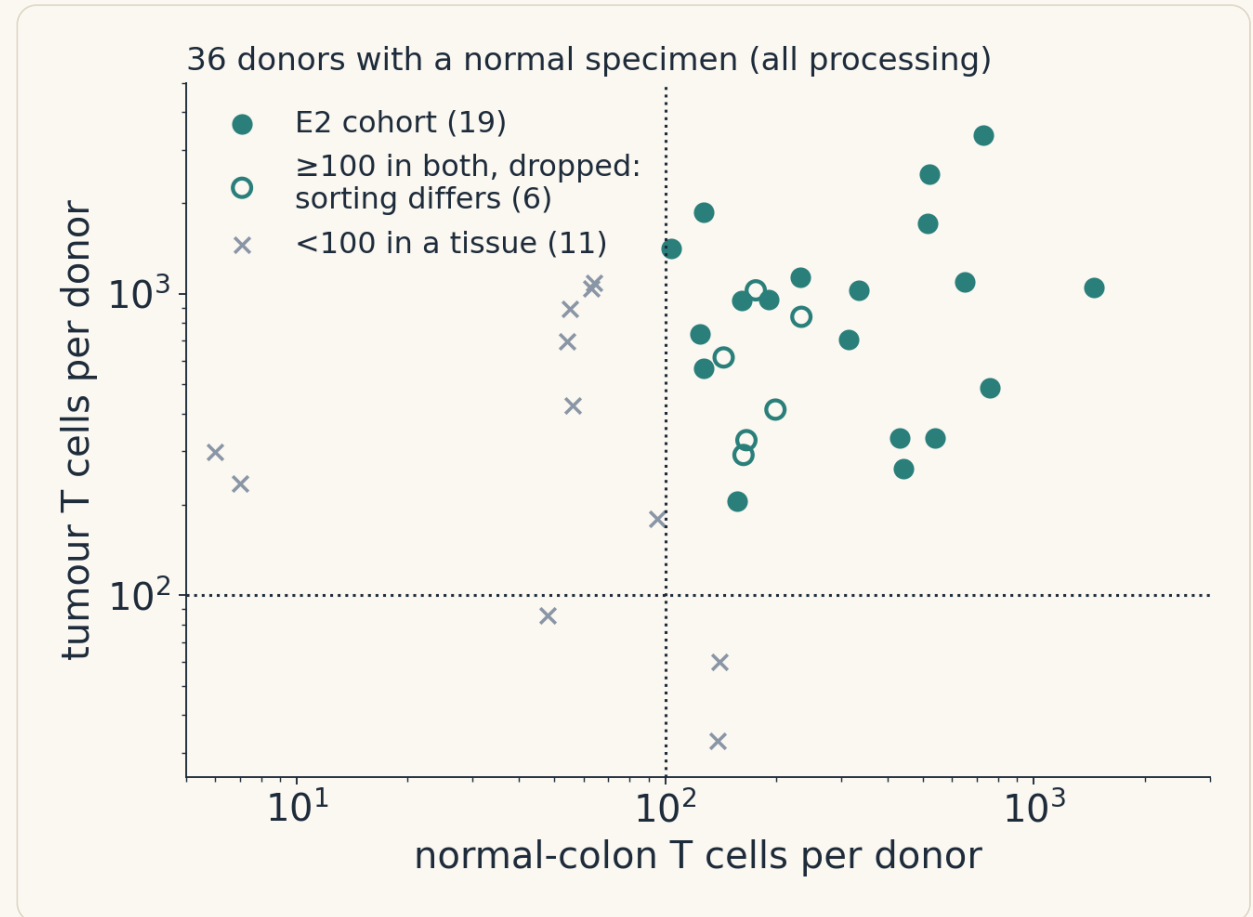
| From: lesson 2 (one donor) and the wording of the August claims.

# External cohorts: one colorectal cohort qualified; the independent lung cohort could not be assessed

**Pelka et al. 2021, colorectal (E2).** 62 patients; 36 with a normal-colon specimen; 25 with  $\geq 100$  CD4/CD8 T cells in both tissues. Restricting both tissues to unsorted cells, because sorting differed within 11 donors, leaves **19**.

**Bischoff et al. 2021, lung (E1).** Data only as a Code Ocean capsule, which refused automated access. ? blocked

*Feasibility from per-cell metadata only; no count data downloaded.*



# Colon study (E2): classifier gate passed, perturbation run complete

✓ registration reviewed

✓ classifier gate

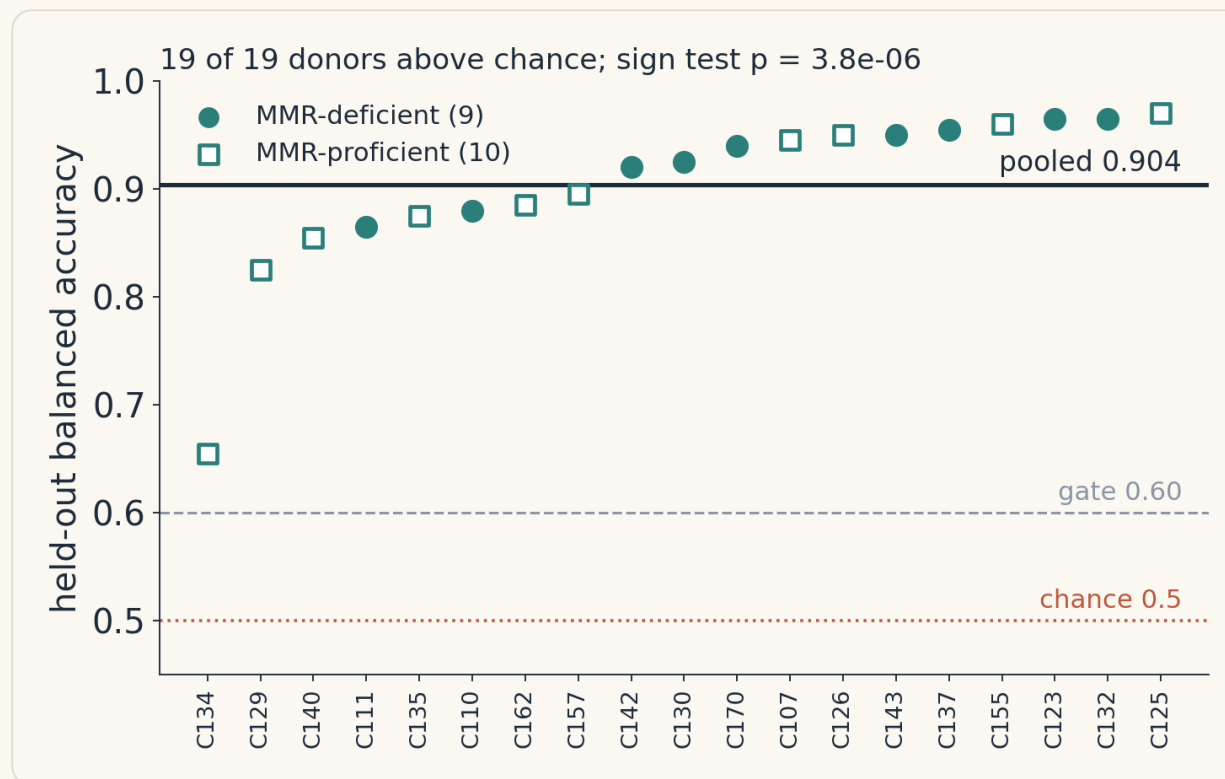
✓ no-op gate

✓ perturbation run complete

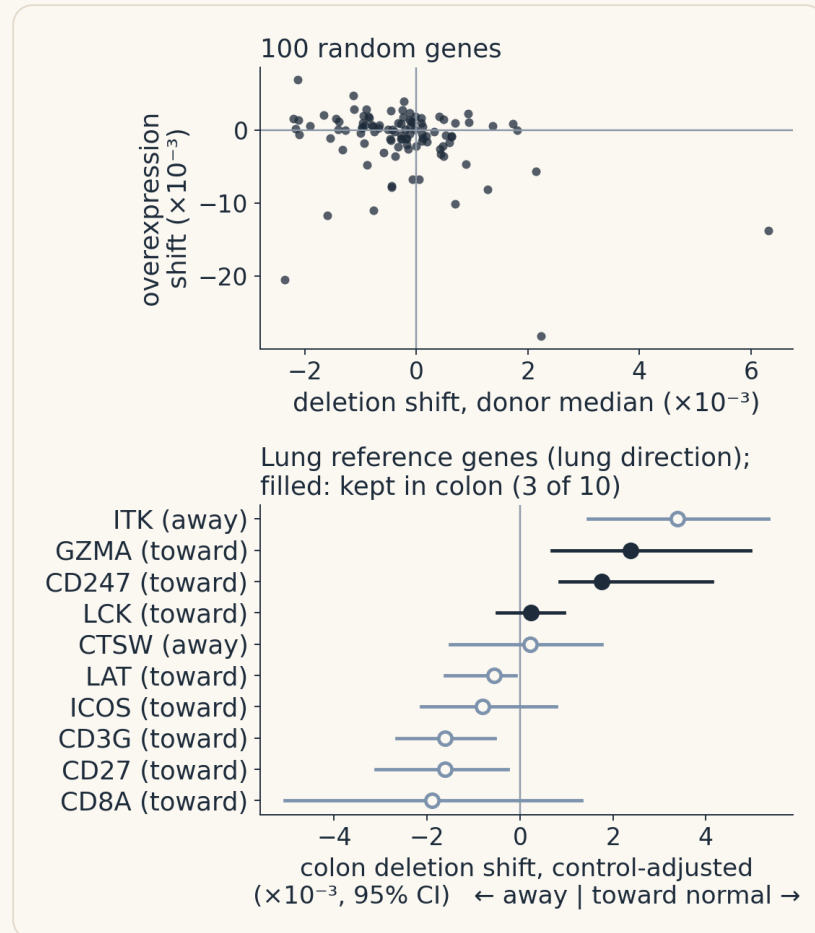
New fold classifiers from the same base model. Pooled held-out balanced accuracy **0.904** (bar 0.60); **19 of 19** donors above chance; lowest 0.655; sign test  $p = 3.8 \times 10^{-6}$ .

Perturbation run: 369 genes  $\times$  19 donors  $\times$  2 arms (14,022 calls), 02:42 JST 1 Oct to 10:20 JST 2 Oct; 32.6 GPU-hours of the 52 authorised. Results on the next slide.

*MMR status is descriptive; no test was registered.*



# Colon results: random genes opposed in the same direction but not stably; the lung gene pattern did not replicate



**Random genes (primary):**  $\rho = -0.245$ , one-sided  $p = 0.0067$  (100,000 permutations); all 100 leave-one-out tests pass; significant in 78% of 10,000 resamples, below the registered 95%.  
control\_draw\_sensitive\_open: direction as in lung, not stable; open.

? open

Lung gave  $-0.593$ . The control genes, a more highly detected set, were also negative (sign-only check).

**Lung reference genes:** 10 testable; 3 keep their lung deletion direction ( $p = 0.95$ ; the bar was 9 of 10). pattern\_not\_replicated. Seven opposite signs are within chance ( $P = 0.17$ ). **X not replicated**

**Panel genes in colon:** 1 of 28 dose-concordant (PRF1, toward normal; it was OPEN in lung). Lung against colon deletion medians, 27 genes:  $p = -0.29$ , compatible with zero.

Readings were fixed before the run. A colon-lung difference is not attributable to tissue alone: study, dissociation, chemistry, annotation, null-gene population and fold models all differ, and 19 donors give less power per gene than 43. New fold models, so the lung models were not tested. Effects are a few thousandths of cosine similarity in both tissues.

# What the results support, and what they do not

## ✓ Supported

- Lung, 43 donors: a fine-tuned classifier separates tumour from normal T cells in held-out donors (0.825; 43 of 43 above chance).
- 13 of 34 testable T-cell genes exceed their matched controls, stably.
- Deletion and overexpression give opposed effects for random genes.
- Colon, 19 donors: the tumour/normal separation is learnable (0.904; 19 of 19).
- Colon: the lung reference-gene pattern did not replicate (3 of 10 kept their lung direction).

## ? Not shown

- That any gene regulates T-cell state. A shift in embedding space is not a mechanism.
- Whether the opposed effects come from the model or from lung tissue: in colon they appear in the same direction but weaker and not stable (open).
- What the classifier reads: T-cell state, ambient RNA or T-cell subset composition.
- Whether the lung models generalise to new lung donors.

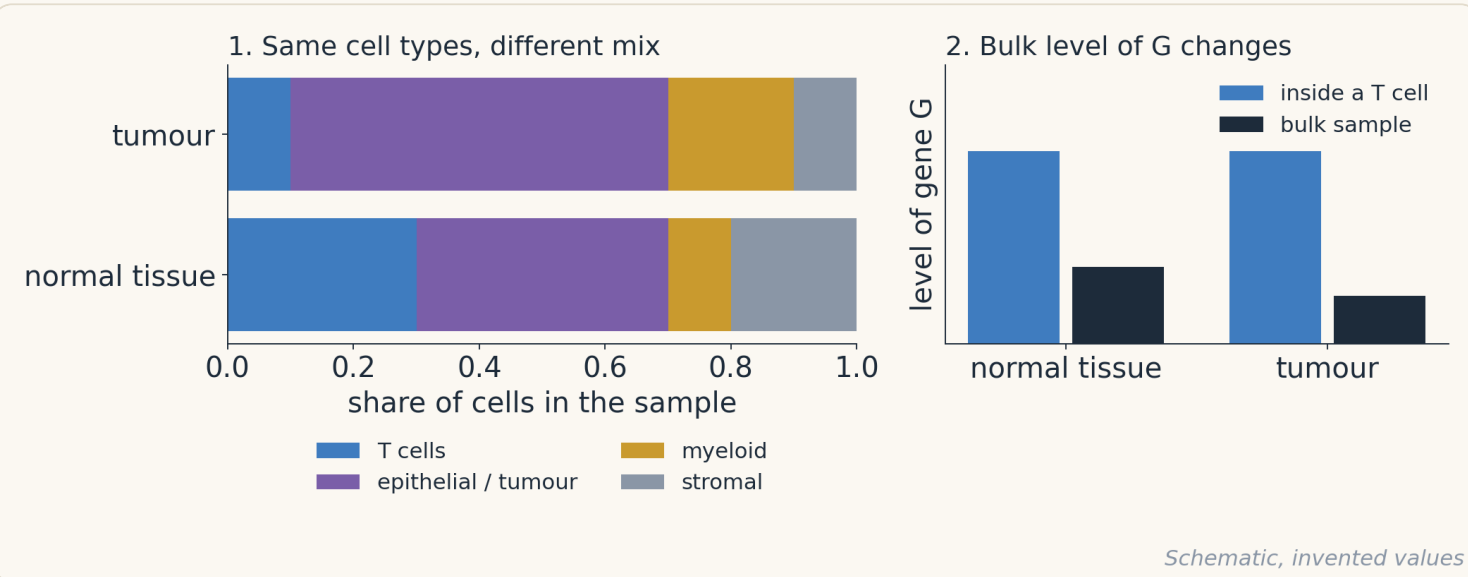
# Bulk RNA-seq: a bulk profile is an average over a mixed population, not a cell

Geneformer was pretrained on single-cell rank encodings. A bulk profile sums many cell types, so a gene's level reflects **composition** as well as **within-cell regulation**.

Any perturbation model fitted on bulk data mixes the two: a shift could mean "this gene changes T-cell state" or "this sample has fewer T cells".

Geneformer itself cannot take a bulk profile as input: the list would be cut at 4,096 tokens and lie outside the pretraining distribution.

? tested twice with a simpler model: next two slides



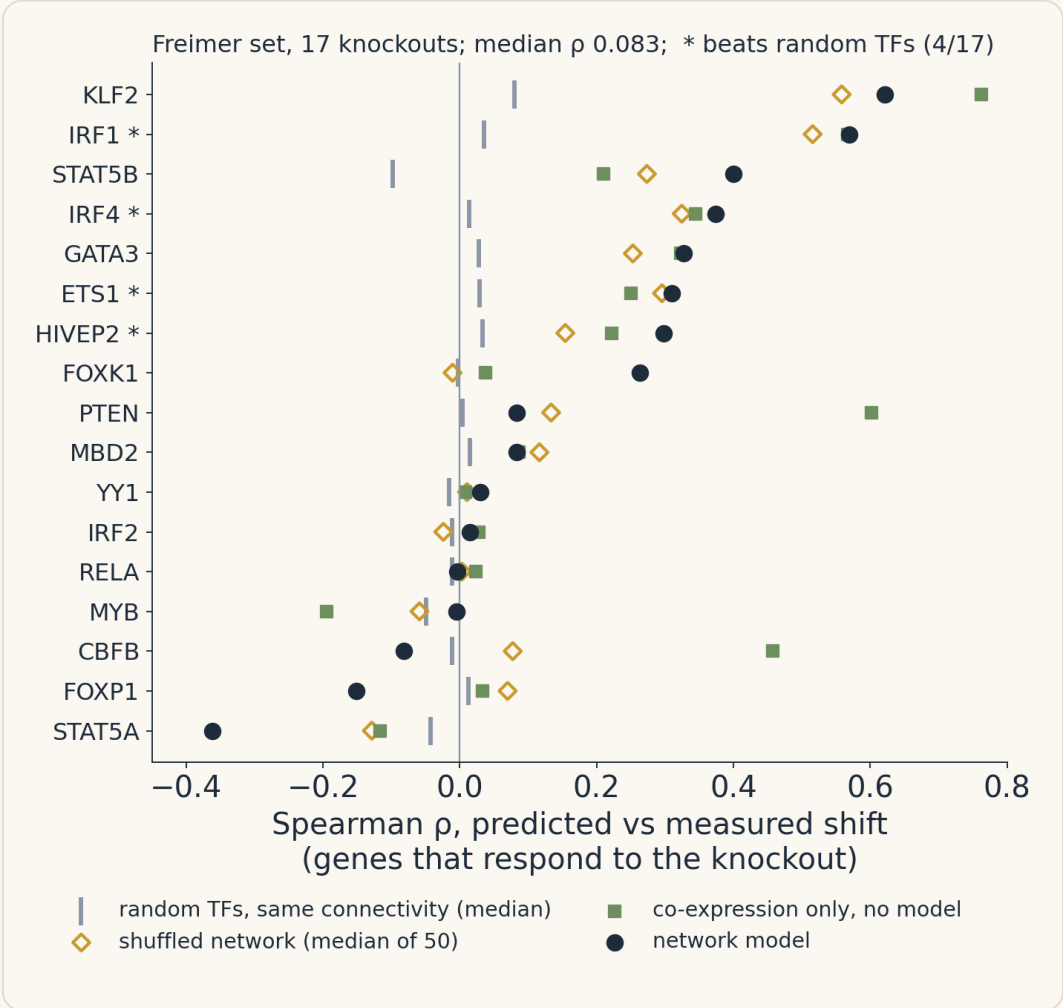
# Bulk tests: a weak, non-specific signal in one knockout set that did not replicate in a second

**Model.** Not Geneformer: a CellOracle-style linear network (ridge regression on transcription-factor-to-target edges of a promoter base network, filtered by matched ATAC-seq), fitted with each knockout left out.

**Ground truth.** Two CRISPR knockout sets in CD4+ T cells, 3 donors each, same laboratory: Freimer et al. 2022 (17 knockouts in the base network) and Weinstock et al. 2024 (38 new ones).

| REGISTERED TEST        | FREIMER, N = 17       | WEINSTOCK, N = 38    |
|------------------------|-----------------------|----------------------|
| Median $\rho > 0$      | 0.083, $p = 0.012$ ✓  | 0.033, $p = 0.21$ ✗  |
| Sign agr. $> 0.5$      | 0.545, $p = 0.013$ ✓  | 0.517, $p = 0.38$ ✗  |
| Beats shuffled network | +0.039, $p = 0.19$ ✗  | +0.035, $p = 0.14$ ✗ |
| Beats random TFs       | 4/17, $p = 0.009$ ✓   | 3/38, $p = 0.30$ ✗   |
| Reading                | RECOVERED_NONSPECIFIC | NOT_RECOVERED        |

By the rule registered before the second run, the first reading does not replicate.



# What the two bulk tests show, and what they leave open

## ✗ Not supported in this design

- Regulator-specific prediction: as a group, the model did not beat a shuffled network (P3 failed in both sets).
- The Freimer signal was matched by co-expression alone (median  $p$  0.21); in Weinstock both fell to chance (0.015).
- Fitted on unperturbed controls only, the network showed no agreement above chance in either set (median  $p$  -0.013 and 0.008).
- Predicted shifts were 55 to 85 times too small (median ratio 0.018 and 0.012).
- Transposable-element arm: the deposited counts had no TE rows; after re-alignment, `TE_RECOVERED_NONSPECIFIC` (slide 34).

## ? Open

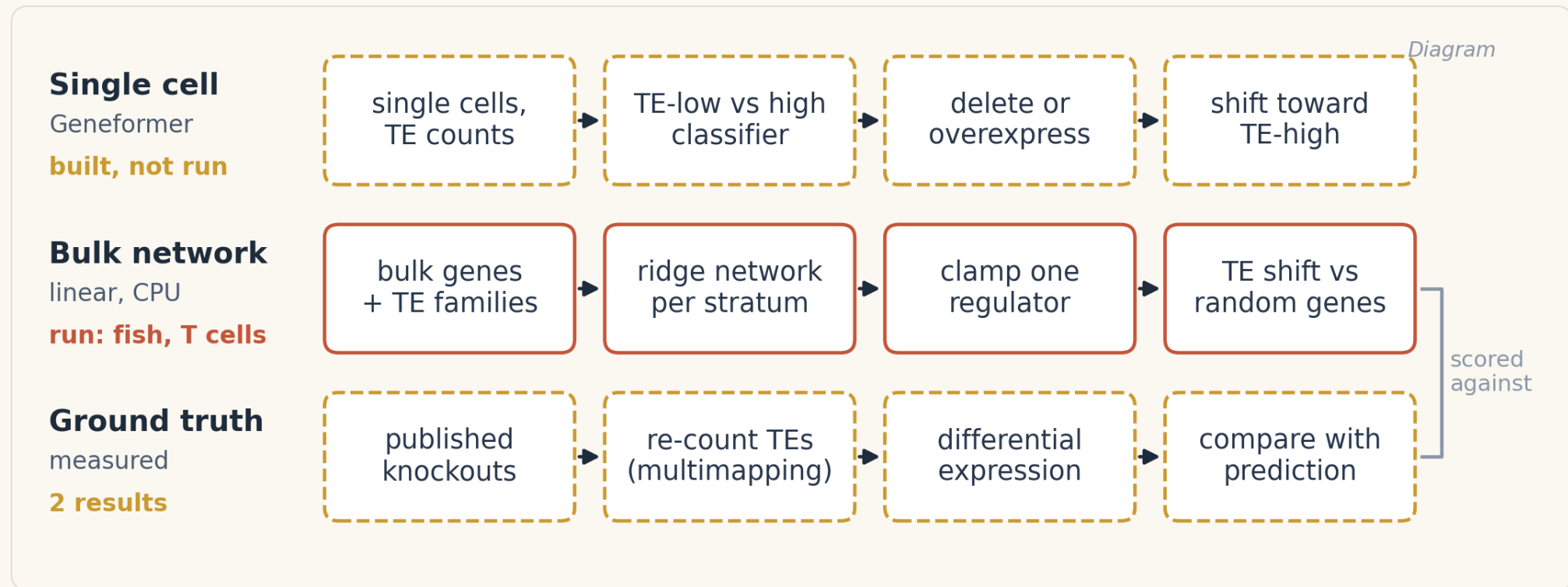
Why the sets differ is untested. Freimer's regulators gave large, overlapping responses (median 394 responsive genes); Weinstock's gave small ones (70.5).

Uncontrolled differences: harvest at 5 days after electroporation versus 8 days after activation; UMI deduplication named in one study only; counting pipeline; unknown donor overlap.

Composition and within-cell regulation cannot be separated in bulk data. A revisit would need a composition covariate registered in advance, or single-cell or sorted data.

# A second project asks which regulators set TE expression

**Transposable elements (TEs)** are repeated sequences that cells keep silenced (KRAB zinc-finger proteins with KAP1, HUSH, DNA and H3K9 methylation). Geneformer\_TE applies the perturbation logic and checks of the T-cell work to TE regulators.



*Bulk results are judged against expression-matched random genes, never against zero.*

# Geneformer has no TE tokens, so the single-cell track reads TE states out through a probe

## Design

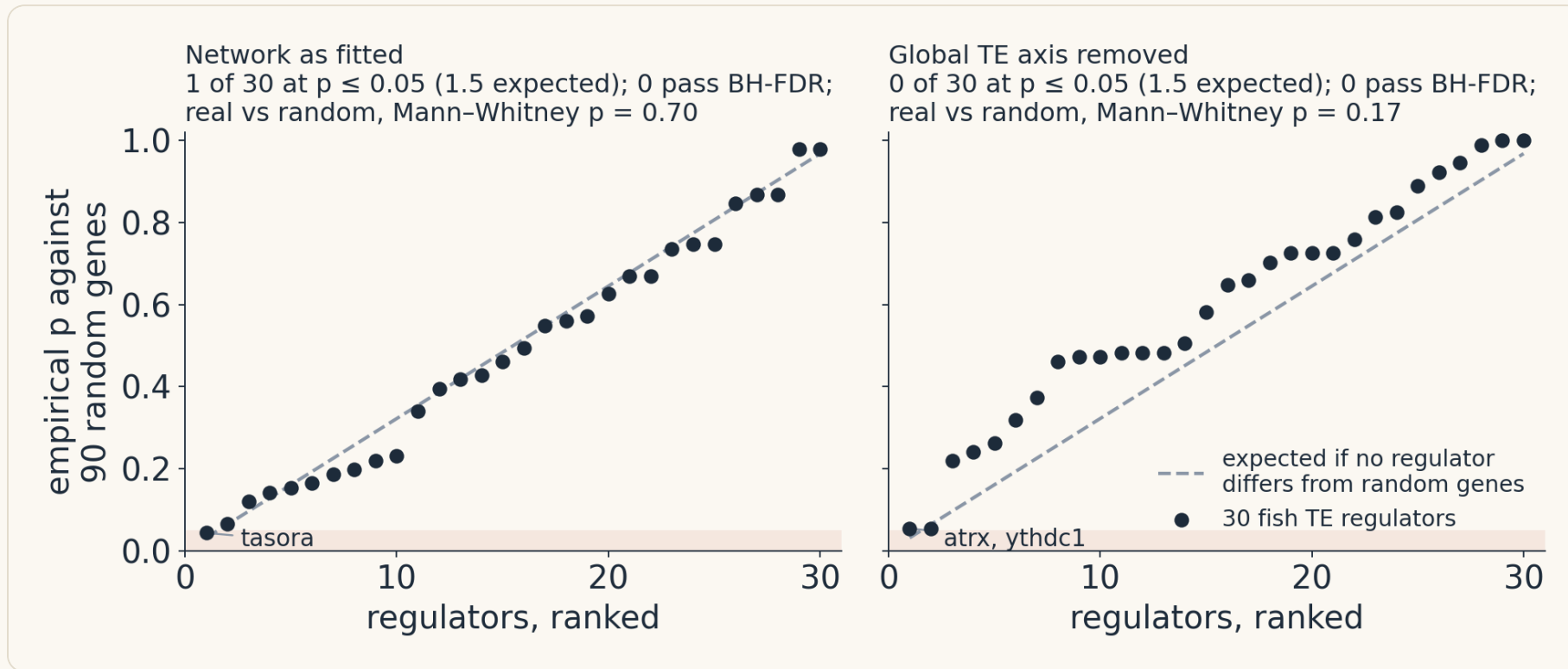
- The vocabulary holds human genes only. TE expression defines the cell states (TE-low, TE-high within each cell type) and is read out by a linear probe from the cell embedding.
- A shift toward TE-high can therefore arise only through genes.
- A classifier gate comes first: donor-level confidence intervals, a sequencing-depth baseline, calibration and per-stratum floors.
- 316M model in bf16, with an fp32 check run on a subset of genes.

## ? built, not run

No human single-cell data with TE counts have been prepared. Candidates: re-process the lung T-cell atlas with a TE counter, or use a public data set with a known TE de-repression.

10x 3' reads are short and often map to many TE copies, so the design stays at the TE family level.

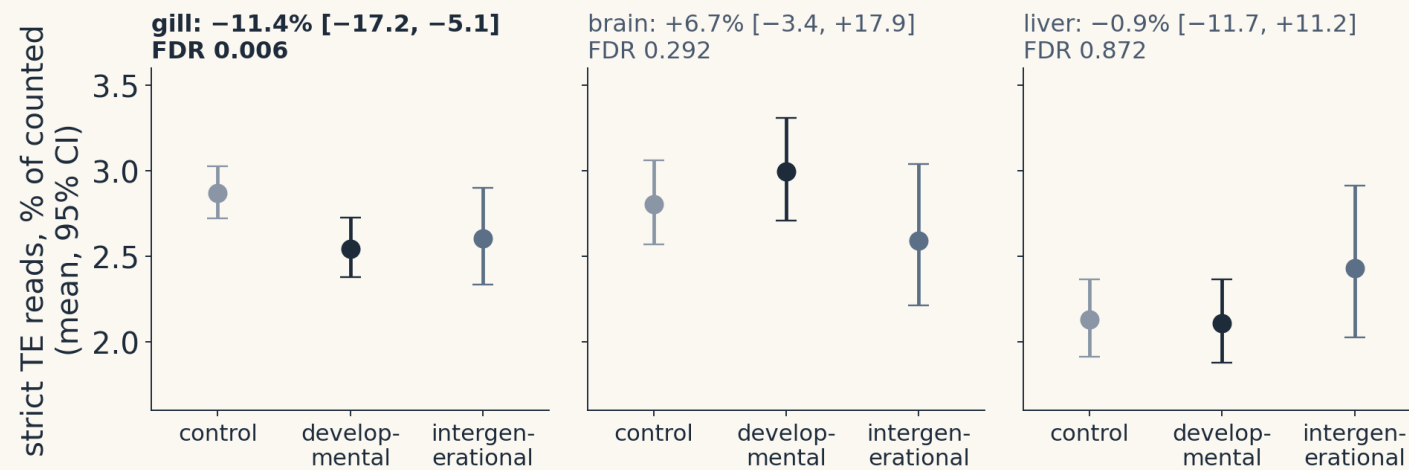
# In 162 fish libraries, the 30 TE regulators did no better than expression-matched random genes



Reef fish (*A. polyacanthus*), 3 tissues  $\times$  3 CO<sub>2</sub> groups; 30 regulators against 90 expression-matched random genes. Without that control, 32,142 of 114,720 tests reached FDR  $< 0.05$ ; with it, none passed, also after removing the global TE axis. **✗ no regulator-specific signal**

# Developmental CO2 lowered the gill TE share; brain and liver showed no robust shift

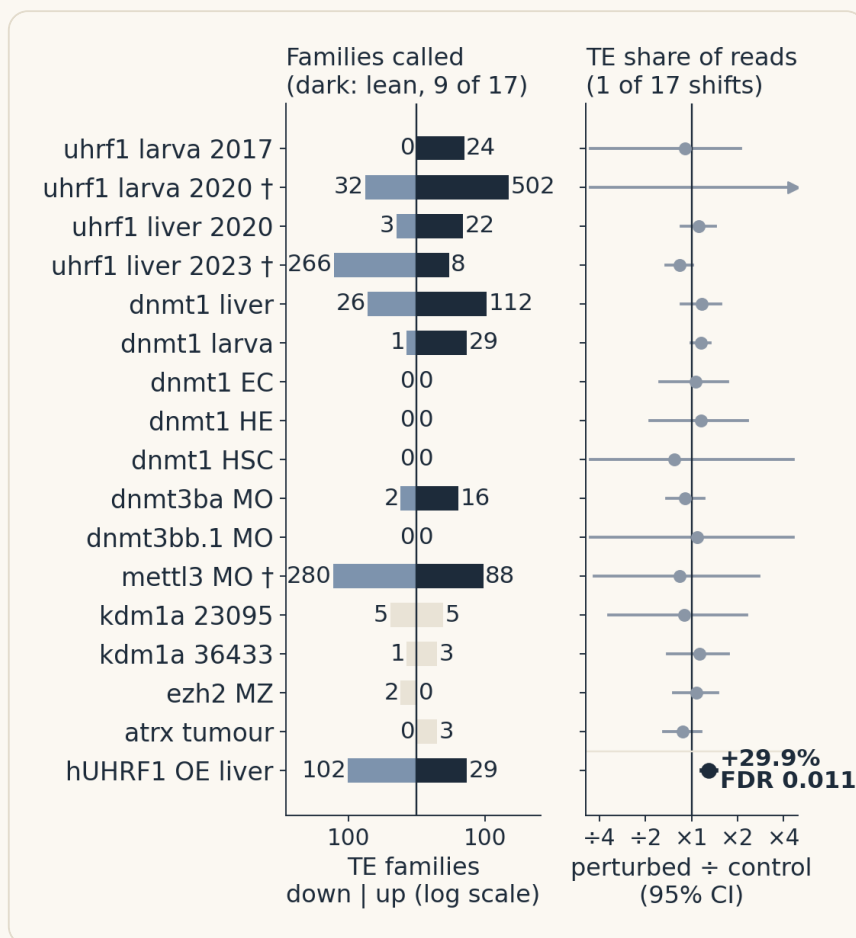
Developmental CO2 vs control, relative change in strict TE share (95% CI)



Gill: 2.87% to 2.55% of counted reads,  $-11.4\%$   $[-17.2, -5.1]$ ,  $p = 0.0009$ , FDR 0.006; the effect persists after QC adjustment and outlier exclusion. Tolerant and sensitive lines did not differ (smallest FDR 0.11).

*Exploratory: not specified in the original study plan. A share of counted reads, not absolute TE expression; a shift in global TE load, not a regulator effect.*

# Zebrabfish ground truth: no TE-regulator loss shifted global TE load detectably; family leans point weakly to derepression



17 published contrasts, 14 GEO series, 112 runs (uhrf1, dnmt1, dnmt3, kdm1a, ezh2, mettl3, atrx); TEs re-counted with one pipeline, each study against its own controls.

**Prespecified, TE share:** 0 of 16 loss-of-function studies shifted at FDR < 0.05. The one shift, human UHRF1 overexpression in liver (+29.9% [16.6, 44.5], FDR 0.011), is mostly compositional: gene-assigned reads fell (−2.55 M per library [−4.98, −0.11],  $p = 0.043$ ), TE reads did not ( $p = 0.86$ ).

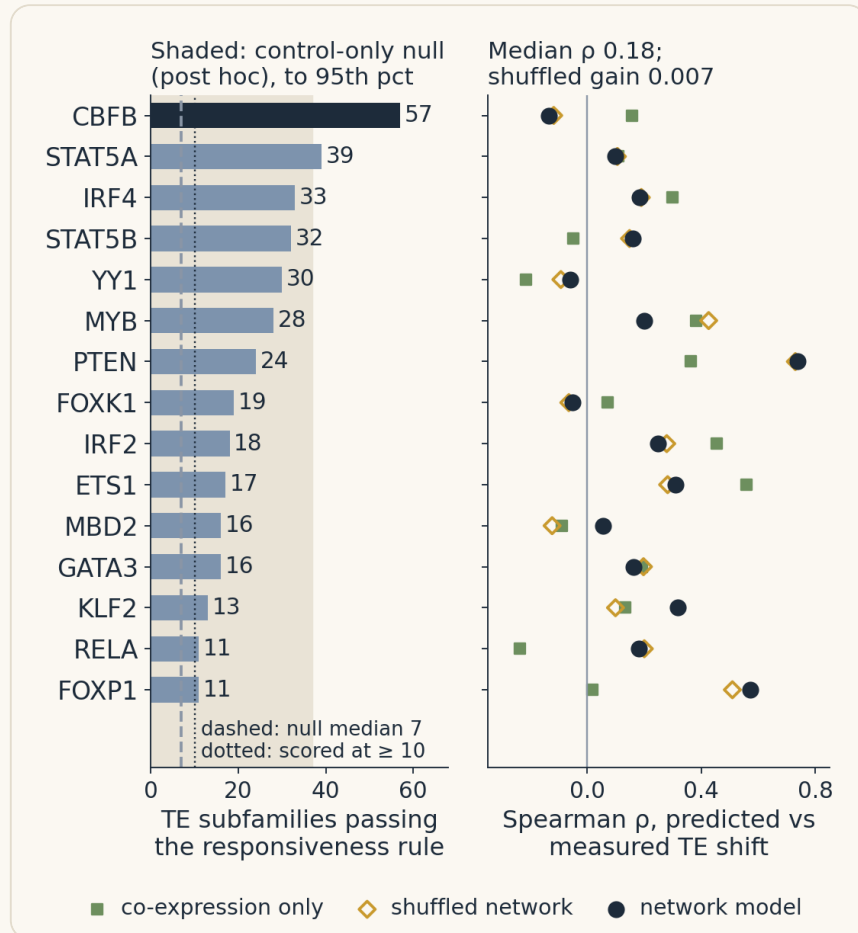
**Exploratory, family leans:** loss-of-function studies with a significant lean leaned up in 6 of 8 (0.75 [0.35, 0.97],  $p = 0.29$ ); for DNA-methylation regulators, 6 of 7 (0.86 [0.42, 1.00],  $p = 0.13$ ). No row passes FDR. dnmt1 liver: 112 families up, 26 down, the gain in LTR families.

**Fish network directions:**  $\rho = -0.18$  against the TE-share change ( $p = 0.50$ ,  $n = 17$ ); family leans agreed in 4 of 9 ( $p = 1.0$ ), at chance level.

**X not supported**

2 to 5 libraries per arm. † Post hoc flag: unique-mapping rate differs by genotype (uhrf1 liver 2023, uhrf1 larva 2020, mettl3 morphant), so their TE direction cannot be separated from library quality. Clutch blocking was added after an interim look and created one upward lean (uhrf1 liver 2020). Cross-species, cross-perturbation test with low power.

# Freimer TE arm: TE shifts mostly within a control-only null (post hoc); the network's TE predictions were non-specific



96 libraries re-aligned with multimappers kept; rules registered before any TE count. Gates passed: gene counts matched the deposited ones (Spearman 0.976 to 0.980, 96 of 96); 712 TE subfamilies expressed.

**Observed:** 15 of 17 representable knockouts had  $\geq 10$  responsive subfamilies (median 19, range 11 to 57). A control-only null (post hoc; one AAVS1 library per donor as pseudo-knockout): median 7, 95th percentile 37, 25% reach 10. Only CBFB exceeds it (57,  $p = 0.010$ ); for genes, 14 of 17 knockouts do.

**Reading** **TE\_RECOVERED\_NONSPECIFIC** ( $n = 15$ ): P1 median  $\rho$  0.18 ( $p = 0.001$ ) and P2 sign 0.64 ( $p = 0.010$ ) pass; P3 gain over shuffled 0.007 ( $p = 0.34$ ) and P4, 1 of 15 beating random TFs ( $p = 0.54$ ), fail. Co-expression alone:  $\rho$  0.13. **X non-specific**

**Host-gene transcription:** 83.3% of TE counts intronic, 4.0% exonic or UTR, 5.5% distal. Distal loci only (S-TE2), the read-through check, left 3 scorable knockouts, too few to test.

**Null result:** the knockouts do not detectably shift TE subfamilies beyond library-to-library variation among controls (CBFB the one possible exception, post hoc). Because P1 and P2 were scored on subfamilies selected at near-chance rates, the TE reading carries less weight than the gene reading. EM-assigned multimappers; 3' tag-seq misses TE transcription away from poly-A sites.

# What the TE work shows so far, and what it does not

## ✓ Supported

- A co-variation network did not meet the registered bar for specific prediction in any controlled test: 162 fish libraries (no regulator beat random genes), Freimer and Weinstock genes (slides 27-28), Freimer TEs (slide 34).
- Without random-gene controls the same run would have reported 32,142 significant regulator-to-family tests.
- Developmental CO2 lowered the gill TE share by 11.4% (exploratory).
- Zebrafish: no loss-of-function study shifted global TE share (0 of 16); fish network directions agreed at chance level (4 of 9).
- Freimer T cells: TE subfamily shifts did not exceed a control-only null, CBFB aside (post hoc).

## ? Not shown

- Any regulator-to-TE effect, in fish or in T cells.
- Whether Geneformer ISP can rank TE regulators: the single-cell track has not run.
- Whether a TE shift in T cells can be separated from host-gene transcription: 83% of TE counts in these 3' libraries are intronic.
- Whether DNA-methylation loss de-represses TEs: 6 of 7 zebrafish leans point up, exploratory and not significant.
- What the global TE axis is biologically: intronic or nascent RNA, or cell composition.

# Summary 1: the evaluation criteria, each traced to evidence

| CRITERION                                                                             | ✓ ADOPTED                                                             | EVIDENCE BEHIND IT                                                  |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------|
|  1   | Register test, direction, threshold, reachable p and wording first    | August ordering read without a plan                                 |
|  2   | Classifier works on held-out paired donors; a no-op gives zero        | July groups came from different studies (0 of 18)                   |
|  3   | Both perturbations on identical cells                                 | 2,424 vs 202 to 1,131 cells; code default 14,738 of 15,179          |
|  4   | Exceed 20 or more matched control genes; flag ambient genes           | S100A8 and S100A9: same direction in 9 of 12; 55 of top 120 ambient |
|  5 | Exceed the random-gene baseline; opposed signs are not evidence       | Random genes: $\rho$ $-0.593$ (100), $-0.608$ (200)                 |
|  6 | Donors as the unit; stable in 95% of resamplings; fixed outcome names | One donor held 74.9% of the SCLC test cells                         |

Status: lab practice (ISP-STD-1, 28 September 2026). Not yet published or tested by other groups.

## Summary 2: what remains to be tested

| STATUS          | QUESTION                                                                  | APPROACH                                                                                                        |
|-----------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| ? open          | Are the opposed effects a property of the model or of lung?               | Colon (E2): same direction, weaker, not stable; lung gene pattern not replicated                                |
| ? blocked       | Do the lung models generalise to new lung donors?                         | E1: the one candidate dataset is not accessible                                                                 |
| ? planned       | Are the opposed effects the same for an unrelated goal?                   | Swap the goal centroid                                                                                          |
| ? planned       | Is the 13-gene result cell state or T-cell subset composition?            | Re-analyse within T-cell subsets                                                                                |
| ? planned       | Can a simpler, interpretable method match the classifier?                 | T-cell program scores (TCAT, Kotliar et al. 2025)                                                               |
| ? not started   | Do results depend on model size (104M vs 316M parameters)?                | Re-run a subset on both sizes                                                                                   |
| ? not started   | Does the method recover genes known to matter?                            | Positive controls; experimental CRISPR screens in T cells                                                       |
| x not supported | Does bulk network ISP recover knockout-specific targets?                  | Two registered tests, 55 knockouts: non-specific, then not replicated                                           |
| x not supported | Did the fish network predict TE responses to regulator loss in zebrafish? | 17 contrasts: $p = -0.18$ , leans agree 4 of 9; loss leaned up 6 of 8, 6 of 7 for DNA methylation (exploratory) |
| x not supported | Does the bulk network predict TE responses to T-cell knockouts?           | Freimer TE arm: non-specific; only CBFB above a control-only null (post hoc)                                    |
| ? not started   | Can Geneformer ISP rank TE regulators in single cells?                    | Needs single-cell data with TE counts                                                                           |

All rows but three are open. The bulk, zebrafish and T-cell TE rows have results: not supported in this design.

# These results could form one methods paper on evaluating in-silico perturbation

*Working title: "Opposite by default: baselines that in-silico perturbation in single-cell foundation models needs"*

**Result.** ✓ established in lung, larger model, our design Random genes give opposed deletion and overexpression effects ( $\rho$   $-0.59$  and  $-0.61$ ).

**Argument.** Gene-level claims therefore need matched-control and random-gene baselines, paired cells, donor-level counting and a registered plan.

? hypotheses The same holds in other tissues, for other goals and model sizes.

**Bulk.** ✗ not supported Across two registered tests (55 knockouts), a bulk network model did not recover regulator-specific effects; its one weak positive was matched by co-expression and did not replicate.

**Fig. 1** Design and the six checks

**Fig. 2** Random-gene baseline in lung

**Fig. 3** Colon, and a swapped goal

**Fig. 4** Four failure modes the checks catch

**Fig. 5** T-cell genes against matched controls

**Fig. 6** Model size, new lung donors, known genes

**Fig. 7** Bulk network ISP against two knockout sets

# Seven tests, three or four questions: where predictions tracked the data, so did the controls

- **Two kinds of test.** Bulk linear network, five tests: four compare predictions with measured perturbations, one is a control test of the fish regulator calls. Geneformer, two tests: perturbation outputs against matched controls and a random-gene null, then a second tissue. No Geneformer prediction has been checked against a measured perturbation.
- **Bulk network, human knockouts.** Freimer RECOVERED\_NONSPECIFIC (gain over a shuffled network +0.039,  $p = 0.19$ ); Weinstock NOT\_RECOVERED (all four tests fail); Freimer TE subfamilies TE\_RECOVERED\_NONSPECIFIC (gain 0.007,  $p = 0.34$ ). As a group the model did not beat a shuffled network in any of the three.
- **Fish regulator calls.** Indistinguishable from expression-matched random genes ( $p = 0.70$ ; 0 of 30 after FDR); their directions did not track zebrafish knockouts ( $p = -0.18$ ,  $p = 0.50$ ; 4 of 9 studies).
- **Geneformer, lung to colon.** The classifier transfers (balanced accuracy 0.825, then 0.904); the deletion/overexpression anti-correlation appears in the same direction but is not stable ( $p = -0.593$ , then  $-0.245$ ; 78% of resamples against a 95% bar); the lung gene pattern did not replicate (3 of 10 keep their sign, within chance).
- **TE biology, exploratory.** Gill TE share lower after developmental CO2 ( $-11.4\%$  [ $-17.2, -5.1$ ]; 54 libraries; not decomposed into TE and gene reads). Zebrafish loss-of-function leans 6 of 8 up ( $p = 0.29$ ), consistent with derepression and not significant.

The seven tests share data and methods and amount to three or four questions. At this power, a negative or open reading leaves room for effects the registered controls could not separate.

# Summary

1. In lung, random genes give opposed deletion and overexpression effects ( $\rho$   $-0.59$  to  $-0.61$ ). Gene-level claims must exceed this baseline.
2. Six evaluation criteria, each traced to a documented error, are now lab practice. In colon, random genes were opposed in the same direction but not stably (open); the lung gene pattern did not replicate.
3. On bulk RNA-seq, a simple network model did not recover regulator-specific knockout effects in two registered tests (55 knockouts); the first test's weak signal did not replicate.
4. For transposable elements, the same network did no better than random genes in 162 fish libraries, and in zebrafish its directions agreed with measured TE responses at chance level; in T-cell knockouts, TE shifts mostly stayed within a control-only null (post hoc; CBFB the exception).

Every number is sourced in SOURCES.md in this talk's folder.

# Glossary

|                                     |                                                                                                       |
|-------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Geneformer</b>                   | A transformer model pretrained on about 104 million single cells, each encoded as a ranked gene list. |
| <b>Token</b>                        | One gene in a cell's ranked list.                                                                     |
| <b>Embedding</b>                    | The model's vector for a cell; similar cells have similar embeddings.                                 |
| <b>Fine-tuning</b>                  | Further training on one task, here tumour versus normal T cells.                                      |
| <b>In-silico perturbation (ISP)</b> | Deleting a gene from, or moving it to the top of, a cell's list and measuring the embedding shift.    |
| <b>Balanced accuracy</b>            | Accuracy averaged over the two classes; 0.5 is chance.                                                |
| <b>Ambient RNA</b>                  | Transcripts from other cells captured during sample preparation.                                      |
| <b>Matched control genes</b>        | Genes with similar detection rate and mean rank to the gene tested.                                   |
| <b>Spearman <math>\rho</math></b>   | Rank correlation, from $-1$ (opposed) through 0 to 1 (aligned).                                       |
| <b>p-value</b>                      | How often chance alone would give a result at least this extreme.                                     |
| <b>Registration</b>                 | Recording the test, threshold and wording before any result exists.                                   |
| <b>GPU-hour</b>                     | One hour of computation on one graphics processor.                                                    |
| <b>Bulk RNA-seq</b>                 | Sequencing of RNA pooled from all cells in a sample.                                                  |
| <b>Pseudo-bulk</b>                  | A bulk-like profile made by summing single-cell data per sample.                                      |
| <b>Gene regulatory network</b>      | A map of which transcription factors may regulate which genes.                                        |
| <b>Transposable element (TE)</b>    | A repeated sequence that can copy itself in the genome; most copies are silenced.                     |